

1 Question (20 points)

Give state diagrams of NFA's with the specified number of states recognizing each of the following languages:

- w contains an even number of 0's, or contains exactly two 1s (6 states)
- $1^*(001^+)^*$ (3 states)

2 Question (20 points)

Give CFG's with the specified number of variables recognizing each of the following languages:

- $A = \{w \mid \text{starts with 0 and has even length, or starts with 1 and has odd length}\}$ (2 variables)
- $A = \{w \mid \text{contains at least two 0's and one 1}\}$ (2 variables)

3 Question (30 points)

Design a TM that recognizes the language $L = \{w_1\$w_2 \mid \text{number of 0's in } w_1 = \text{number of 1's in } w_2; \text{ number of 1's in } w_1 = \text{number of 0's in } w_2\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

4 Question (30 points)

Let a k -PDA be a pushdown automaton that has k stacks. Thus a 0-PDA is an NFA, 1-PDA is conventional PDA. Give an example language for which 2-PDA recognizes that language but 1-PDA can not, prove it.